Homework 4
Elementary Linear Algebra
(MATH 2250-03), Spring, 2018

Check: 03/01/2019 Due: 03/04/2019

Section 2.2 - The Inverse of a Matrix

Let $A = \begin{bmatrix} 3 & 2 \\ 7 & 4 \end{bmatrix}$, let $B = \begin{bmatrix} 3 & -4 \\ 7 & -8 \end{bmatrix}$, and let $C = \begin{bmatrix} 1 & 0 & -2 \\ -3 & 1 & 4 \\ 2 & -3 & 4 \end{bmatrix}$.

1. Compute $\left(AB\right)^{-1}$.

2. Compute $\left(C^T\right)^{-1}$.

Section 2.3 - Characterizations of Invertible Matrices

3. Let A and B be $n \times n$ matrices and suppose $BA = I$. Show that A is invertible and that $B = A^{-1}$.

4. Let A and B be $n \times n$ matrices and suppose AB is invertible. Show that A is invertible and give a specific formula for the inverse of A in terms of $(AB)^{-1}$ and B.
(HINT: Use what you showed in problem 3.)

Section 2.4 - Partitioned Matrices

5. Suppose A, B, and C are $n \times n$ matrices, and let X, Y, and Z be $n \times n$ matrices such that

$$\begin{bmatrix} X & 0 \\ Y & Z \end{bmatrix} \begin{bmatrix} A & 0 \\ B & C \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}.$$

Find formulas for X, Y, and Z in terms of A, B, and C.
(HINT: Use what you showed in problem 3.)

Section 2.5 - Matrix Factorizations

6. Let $A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix}$ and let $b = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$. Solve the equation $Ax = b$ by using the following LU factorization of A:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$